Akoizo Original Practice - Form 1 - Answer key

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) How is the final score calculated?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

A. Sum of the three best flight scores

B. Best single flight

C. Average of all flights

D. Sum of all five flights

Correct: A

2. (2 points) Which launch method is NOT allowed?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

A. Banking the launch

B. Launching from the floor

C. Standing on a chair to launch

D. Using a 90 cm handle

Correct: C

3. (2 points) A glider repeatedly stalls in the glide. What may help?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

A. Add decalage

B. Launch harder

C. Move the CG forward slightly

D. Move the CG back

Correct: C

4. (2 points) Why do tight circles shorten glides?

Unit: Glide physics: sink rate and glide ratio.
Worked-case data: A glider reaches 9.0 m and glides at 3.2 m/s with glide ratio 8. Estimate glide time.
Separate investigation observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

A. They reduce drag

B. They slow the glider

C. Banking more requires more lift, increasing sink rate

D. They increase glide ratio

Correct: C

5. (2 points) Why start official flights early in the period?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

A. To leave time for spare flights in case some are poor

B. Early flights score double

C. Late flights are not timed

D. It is required

Correct: A

6. (4 points) Why build under the target mass and add ballast?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

answer: Ballast lets teams fine-tune both total mass and CG, and adjust for scale differences. (4 points).

7. (4 points) Why should elastic relaxed length be logged?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

answer: Elastic length and condition change stored energy, so logging it links launch energy to flight results. (4 points).

8. (4 points) What is the benefit of declaring a trim flight?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

answer: It lets the team test trim or the room without using one of the five official flights. (4 points).

9. (4 points) What safety practices apply to the launch handle?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

answer: One student holds it, it is under 1 m, the elastic stays attached, and fingers and faces stay clear. (4 points).

10. (4 points) Why change one trim setting at a time?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

answer: So the effect of each change is clear and can be reversed if it makes things worse. (4 points).

11. (6 points) Solve this case and explain your reasoning: A glider reaches 9.0 m and glides at 3.2 m/s with glide ratio 8. Estimate glide time.

reasoning: Sink rate = airspeed ÷ glide ratio = 3.2 ÷ 8 = 0.40 m/s.
Glide time = 9.0 m ÷ 0.40 m/s = 22.5 s.
Add the few seconds of climb for total flight time. (3 points).

conclusion: About 22.5 s of glide, plus the climb. (3 points).

12. (6 points) Solve this case and explain your reasoning: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?

reasoning: Current best three: 23 + 21 + 8 = 52 s.
Two more flights near 22 s would replace the 8 s and add a tiebreaker.
Adjust to avoid the beam, then fly both remaining official flights. (3 points).

conclusion: Fix the beam problem, then use both spare flights to replace the 8 s flight. (3 points).

13. (2 points) What should the team check first?

Supplied synthetic observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

A. Wing and tail alignment after the hard landing

B. The flight log

C. The room’s ceiling

Correct: A

14. (2 points) Define “Elastic potential energy” in the context of Elastic Launched Gliders.

answer: ½kx^2 for an ideal spring. (2 points).

15. (2 points) Define “Official flight” in the context of Elastic Launched Gliders.

answer: Declared scoring flight; up to five per team. (2 points).

16. (2 points) Define “Ceiling hit” in the context of Elastic Launched Gliders.

answer: Contact with the ceiling that wastes height. (2 points).

17. (2 points) Define “Decalage” in the context of Elastic Launched Gliders.

answer: Wing incidence minus stabilizer incidence. (2 points).

18. (2 points) Define “Wing loading” in the context of Elastic Launched Gliders.

answer: Weight divided by wing area. (2 points).